# Applied Economics

## Causality vs. Correlation

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# Announcement

Homework 2 will be uploaded this Friday and is due on **Monday, November 26 at 6 pm.**

# Plan for this Lecture

1. Correlation vs. Causality
2. Endogeneity
    - Reverse Causality
    - Measurement Error
    - Omitted Variable Bias
3. Model Specification

# Correlation vs. Causality

# Causality

We say $X$ causes $Y$ if:

When we change the value of $X$ **without changing anything else**, then $Y$ would also change as a result.

An ideal way to identify causal effects are **Randomized Control Trials** (RCTs)

- you randomly assign people to a treatment and a control group.
  $\rightarrow$ the treatment assignment is independent of all other potential factors that have an impact on $Y$.
  $\rightarrow$ treatment effects can be estimated by simply comparing means across groups
- example: assign students to small vs. regular classes to identify the impact of class size on test performance

# Causal Analysis with Observational Data

RCTs are costly and often face ethical issues.
Also, it is not always clear to what extent results from a (small) RCT can be generalized to other contexts (e.g. other countries).

Thus, we often use observational data (as opposed to experimental data).
With this type of data, causal inference is usually more difficult.

Coefficients in a (multiple) linear regressions measure (partial) correlations between variables. They can, but they don't have to reflect causal effects!

# Correlation vs. Causality: Example

Regress height of son ($S_i$) on the height of his father ($F_i$), after standardizing both variables ($\rightarrow$ mean = 0, sd = var = 1): $S_i = \beta_0 + \beta_1 F_i + u_i$

- $\beta_1$ might plausibly reflect a causal effect of paternal height on height of the son.
- $\hat{\beta}_1 = \frac{\widehat{cov}(S_i, F_i)}{\widehat{var}(F_i)} = \widehat{cov}(S_i, F_i) = \widehat{cor}(S_i, F_i)$

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We can flip dependent and independent variable: $F_i = \delta_0 + \delta_1 S_i + e_i$

- $\delta_1$ cannot plausibly reflect a causal effect
- $\hat{\delta}_1 = \frac{\widehat{cov}(F_i, S_i)}{\widehat{var}(S_i)} = \widehat{cov}(F_i, S_i) = \widehat{cor}(F_i, S_i) = \hat{\beta}_1$

# Correlation vs. Causation: Examples

**Take-away:**

Neither estimated nor true slope parameters necessarily reflect a causal effect.

Regressions per se are just tools to analyse (partial) correlations between variables.

Causality is an assumption that we add to it, and that needs to be based on economic theory/common sense.

# Correlation vs. Causation: More examples

Even if we have a causal model in the background, there are several reasons why correlations in observational data might not reflect the causal effect of interest:

- the observed correlation between *education* and the *wage* might not reflect the causal effect of education, because individuals who choose to get more education might differ from those who don't (higher innate ability, motivation) [omitted variables]
- the observed correlation between the *minimum wage level* and the *employment rate* might not reflect the causal effect of the minimum wage because policymakers may increase the minimum wage in times when the employment rate is high [reverse causality/simultaneity]

# Endogeneity

# When do partial correlations reflect causal effects?

MLR.4 is central to obtain causal effects. $E(u_i|\boldsymbol{X}_i) = 0 \Leftrightarrow E(u_i|\boldsymbol{X}_i) = E(u_i)$

- $u_i$ is in expectation independent from the regressors.
- The regressors behave almost as in a randomized experiment.

Failure of this assumption is often called *endogeneity*:
Correlation between one or more regressors and the error term

Implication: OLS estimator is biased and inconsistent:

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\boldsymbol{X'X})^{-1}(\boldsymbol{X'u}) \xrightarrow{p} \boldsymbol{\beta} + E[\mathrm{X_i X_i'}]^{-1} \underbrace{E(\mathrm{X_i} u_i)}_{\neq 0} \neq \boldsymbol{\beta}$$

$$E[\hat{\boldsymbol{\beta}}] = \boldsymbol{\beta} + E[(\boldsymbol{X'X})^{-1}(\boldsymbol{X'u})] \neq \boldsymbol{\beta}$$

# Endogeneity in the Simple Regression Model

$Y_i = \beta_0 + \beta_1 X_i + u_i,\ i = 1, ..., n$

$$\hat{\beta}_1 = \beta_1 + \frac{\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})u_i}{\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2}$$

Under MLR.1 to MLR.3 $\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})u_i \xrightarrow{p} E[(X_i - \mu_X)u_i] = cov(X_i, u_i)$

$\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2 \xrightarrow{p} E[(X_i - \mu_X)^2] = var(X_i) = \sigma_X^2$

Thus $plim(\hat{\beta}_1) = \beta_1 + \frac{cov(X,u)}{\sigma_x^2}$

$\Rightarrow$ **Direction of the inconsistency depends on the sign of** $cov(X, u)$**.**

# Sources of Endogeneity

Frequent reasons for endogeneity are

- reverse causality
- measurement error in the explanatory variable
- omitted variables

# Reverse Causality

We talk about reverse causality when $Y$ causes $X$ **and** $X$ causes $Y$.

**Example:**

- We are interested in the effect of the police force on crime:
  $crime_i = \beta_0 + \beta_1 police\ funding_i + u_i$
- But the crime rate also affects government spending on police:
  $police\ funding_i = \delta_0 + \delta_1 crime_i + v_i$
- Assume: $cov(u_i, v_i) = 0$
- Still, we get endogeneity: $cov(police\ funding_i, u_i) \neq 0$,
  because $pol\ funding_i$ (linearly) depends on $crime_i$ and $crime_i$ depends on $u_i$

# (Classical) Measurement Error in the Regressor

Regression model of Interest: $Y_i = \beta_0 + \beta_1 X_{1i} + u_i$ with $E[u_i|X_{1i}] = 0$

We do not observe $X_{1i}$ but just a noisy measure $\widetilde{X}_{1i} = X_{1i} + w_i$
with $E[w_i] = 0$, $E[w_i X_{1i}] = 0$ and $E[w_i u_i] = 0$

We thus consider the following regression model: $Y_i = \beta_0 + \beta_1 \widetilde{X}_{1i} + e_i$
with $e_i = \beta_1 X_{1i} - \beta_1 \widetilde{X}_{1i} + u_i = -\beta_1 w_i + u_i$

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In this model: $Cov(e_i, \widetilde{X}_{1i}) \neq 0$

$$\begin{aligned} Cov(e_i, \widetilde{X}_{1i}) =& E(e_i \widetilde{X}_{1i}) \\ =& E[(-\beta_1 w_i + u_i)\widetilde{X}_{1i}] \\ =& -\beta_1 E[w_i \widetilde{X}_{1i}] + E[u_i \widetilde{X}_{1i}] \\ =& -\beta_1 E[w_i X_{1i} + w_i^2] + E[u_i X_{1i} + u_i w_i] \\ =& -\beta_1 E[w_i^2] \end{aligned}$$

# Measurement Error causes attenuation bias

Thus

$$
\begin{aligned}
plim(\hat{\beta}_1) &= \beta_1 + \frac{Cov(e_i, \widetilde{X}_{1i})}{Var(\widetilde{X}_{1i})} \\
&= \beta_1 - \beta_1 \frac{E[w_i^2]}{Var(X_{1i} + w_i)} \\
&= \beta_1 - \beta_1 \frac{Var(w_i)}{Var(X_{1i}) + Var(w_i)} \\
&= \beta_1 \frac{Var(X_{1i}) + Var(w_i) \breve{} Var(w_i)}{Var(X_{1i}) + Var(w_i)} \\
&= \beta_1 \frac{Var(X_{1i})}{Var(X_{1i}) + Var(w_i)}
\end{aligned}
$$

$0 < \frac{Var(X_{1i})}{Var(X_{1i})+Var(w_i)} \leq 1 \Rightarrow \hat{\beta}_1$ biased towards zero (attenuation bias)

# Omitted Variables

Relevant regression model: $Y_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + u_i$ with $E[u_i|X_i, Z_i] = 0$

Estimated regression model: $Y_i = \beta_0 + \beta_1 X_i + v_i$ with $v_i = \beta_2 Z_i + u_i$

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In this model:

$$\begin{aligned} Cov(X_i, v_i) &= E[X_i v_i] \\ &= E[X_i \beta_2 Z_i + X_i u_i] \\ &= \beta_2 E[X_i Z_i] \end{aligned}$$

# Omitted Variables

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The OLS-estimator for $\beta_1$ will be inconsistent and biased, if

- $\beta_2 \neq 0$: **The omitted variable $Z_i$ is relevant for outcome $Y_i$.**
- $corr(X_i, Z_i) \neq 0$: **The omitted variable is correlated with the included regressor $X_i$.**

# Omitted Variables: Direction of the Bias

Thus

$$plim(\hat{\beta}_1) = \beta_1 + \frac{Cov(v_i, X_i)}{Var(X_i)}$$
$$= \beta_1 + \beta_2 \frac{E[X_i Z_i]}{Var(X_i)}$$

Direction of the **Omitted Variable Bias** depends on

- the sign of $\beta_2$ (impact of the omitted var. on $Y$) and
- the direction of $Cov(Z_i, X_i)$ (correlation of omitted var. and included regressor).

# OVB: Example

# Omitted Variables: Example

Our aim is to estimate the causal effect of education on the wage.

Long model: $log(wage)_i = \beta_0 + \beta_1 education_i + \beta_2 IQ_i + u_i$

Since we typically do not observe results of an IQ-test, we estimate:

$log(wage_i) = \beta_0 + \beta_1 education_i + v_i$

with $cov(education_i, v_i) = \beta_2 \cdot E[education_i \cdot IQ_i]$

# Omitted Variables: Example

«message = F»= df = read.csv("wage2.csv") library(sandwich) library(lmtest) $model_short <$
$-lm(log(wage)\ educ, data = df)coeftest(model_short, vcov. = vcovHC(model_short, type =$
$"HC1"))@According to our estimate, an additional year of education raises the wage by 6\%.$

# Omitted Variables: Example

In this data set, we do have results of an IQ test for all individuals and can thus estimate the long model. «message = F»= $model_long <- lm(log(wage)\ educ + IQ, data = df)coeftest(model_long, vcov. = vcovHC(model_long, type = "HC1"))@According to our estimate, an additional year of education raises the wage by 4\%.$

# Omitted Variables: Example

In the short model, $\hat{\beta}_1$ is **biased upward** because

- $\beta_2 > 0$: *IQ* has a positive impact on *log*(*wage*)
- $cov(IQ_i, educ_i) > 0$: *education* is positively correlated with the omitted variable
  «message = F»= cor(df*educ*, *df*IQ) @

Thus, we *overestimate* the true positive impact of education on wages; part of the correlation reflects the *indirect* effect that individuals with higher education have a higher IQ, and therefore have higher wages.

In real-world applications, we do not have data on the omitted variable.
We can only speculate about its correlation with the included regressor(s) and its impact on the dependent variable, to think about how its omission affects the parameter of interest.

# Model Specification

# How to choose which regressors to include in your model?

If our aim is to model causal relationships, the choice of which regresssors to include should

- be based primarily on economic arguments
- aim to make MLR.4 plausible
- **not** be based primarily on statistical tools ($R^2$, adjusted $R^2$, $t$-tests, $F$-tests.)

If our aim is prediction rather than causal analysis, statistical tools can be very helpful.
Aim: explain $Y$ as good as possible

# Choice of regressors in causal analysis

We are typically interested in the causal effect of one specific variable $X_1$.
What other variables $X_j$, $j = 2, \ldots, k$ should we add as control variables?

- Good control variables: **help to avoid OVB**
- Bad control variables: **mediator variables** (could be considered as an outcome variable themselves)

# Example for bad control variable

Suppose, we are interested in estimating the causal effect of agricultural pesticide use on health care expenditures.

$$\textit{Health Expenditures}_r = \beta_0 + \beta_1 \textit{pesticides}_r + \ldots + u_i,\ r = 1, \ldots, R \text{ (region)}$$

Should we control for the number of doctor visits made?

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Suppose, we are interested in estimating the causal effect of agricultural pesticide use on health care expenditures.

$$\textit{Health Expenditures}_r = \beta_0 + \beta_1 \textit{pesticides}_r + \ldots + u_i,\ r = 1, \ldots, R \text{ (region)}$$

Should we control for the number of doctor visits made?

No! An increase in the number of doctor visits is a **channel** through which pesticide use increases health care expenditures.

If we control for the number of doctor visits, $\beta_1$ would reflect the impact of pesticide use on health care expenditures that are not generated by more doctor visits $\rightarrow$ a parameter that is generally not of pratical interest.

# Recap and Outlook

# Recap

- We are often interested in estimating causal relationships between economic variables.
- This requires that we interpret the linear model as some stylized (economic) model, where the $\beta$s describe causal effects of $\boldsymbol{X}$ on $Y$.
- When we interpret coefficients in that way, the mean independence assumption MLR.4 is crucial to get consistent estimates of the causal effects with OLS.
- OLS estimators consistently estimate the **(partial) correlation** between $X_j$ and $Y$. But if there is **endogeneity**, this does not represent the underlying causal relationship.
- Endogeneity can arise because of reverse causality, measurement error and omitted variables. It is an extremely frequent problem in the social sciences.
- When choosing which control variables to include, our aim is to make MLR.4 plausible, and to avoid inclusion of mediator variables.

# Outlook

How can we address the endogeneity issue?

**Instrumental variable estimation** can be used to address omitted variables, measurement error and reverse causality.

**Panel data** [data on cross-sectional units for multiple points in time] can also be used to address many forms of OVB, as it allows to control for all unobservable factors that are time-invariant.

# APPENDIX

# What do slope coefficients in a MLR measure? The Frisch-Waugh Theorem

In a MLR, $\hat{\beta}_j$ reflects the partial correlation between $X_j$ and $Y$, after the effects of other regressors have been taken out.

$$Y_i = \beta_0 + \beta_1 X_{1i} + \boldsymbol{X}'_{2i}\boldsymbol{\beta}_2 + u_i, i = 1, \ldots, n$$

1. Regress $X_{1i}$ on $\boldsymbol{X}_{2i}$: yields residuals $\hat{e}_{x_1 i}$
2. Regress $Y_i$ on $\boldsymbol{X}_{2i}$: yields residuals $\hat{e}_{yi}$
3. Regress $\hat{e}_{yi}$ on $\hat{e}_{xi}$

$\rightarrow$ estimated slope parameter in step 3 is identical to the OLS estimator $\hat{\beta}_1$ in the MLR! (Frisch-Waugh Theorem)

# The Frisch-Waugh Theorem

The error $\hat{e}_{x_1 i}$ can be thought of as the component (or variation) of $X_{1i}$ that is not linearly explained by $\boldsymbol{X}_{2i}$.

Thus, the coefficient $\beta_1$ is a (rescaled) measure of linear correlation of $X_{1i}$ and $Y_i$, after **partialling out (controlling linearly for) the effects of all variables in $\boldsymbol{X}_{2i}$**

$\rightarrow$ Basis for the **ceteris paribus interpretation** of the model coefficients in a multivariate regression set-up.

But: we keep only the other included regressors fixed. Causal inference requires that there are no components left in the error that are systematically correlated with $X_1$.